InkProf | v1.3.0 | 2026-10-25

Keep your i1Pro 2.
Take control of your prints.

Find out what your printer, paper and ink can reproduce. Build a profile, check the result and show your client what you deliver.



Spectral measurement of a printed colour target - the basis for profiling and verification.

Keep your instrument - keep building profiles

InkProf uses ArgyllCMS to communicate with the i1Pro 2. You are independent of i1Profiler. Keep using the instrument you already own.

Know what your image can achieve on your paper

An ICC profile describes your printer, paper and ink combination. It supports soft proofing and print verification. Make better-informed image and paper decisions before delivery. [1, 2]

Make the next print easier to get right

The project saves settings, measurements and revisions. Trace what produced the result and investigate changes. This can reduce unnecessary test prints and save time, paper and ink. Save completed measurement rows and continue later from the next unfinished row.

Give your client concrete evidence

The certificate shows reference colours, measurements and reviewer decisions. Discuss quality using shared evidence. Explain what was checked and what the client can expect. [3]

Your existing instrument. A profile for your print system. A result you can check and document. InkProf is open source and free to download; MATLAB requires a licence.

EVERYDAY BENEFITS

More control. Less guesswork.

Make repeat work easier to follow

When a client orders more prints or adds to a series, you want to know which conditions produced the earlier result. InkProf keeps materials, settings, measurements and iterations together in one project. This makes it easier to reproduce the process and spot changes that need investigating.

Open source means transparency

InkProf is not a closed "black box". Every step, from colour patches to measurement, calculation and documented results, can be inspected. You can follow how difficult colours are handled, right up to the limits of your printer, paper and ink. The code and saved evidence help you investigate differences and make informed decisions.

Describe quality with measurements

Measurement gives photographers and print businesses a concrete language for discussing quality. InkProf reports the difference between the intended colour, the profile's prediction and the measured print. Standards references provide context. The measurement certificate records the scope, results and decision.

A connected colour workflow

InkProf handles printer profiling and its documentation. Combine it with a calibrated and profiled display, correct colour management when printing and suitable viewing light. A profile applies to one specific combination of printer, paper, ink and print settings. [1, 2]

Sources behind these benefits

Web sources checked on 2026-10-03. Adobe describes predictable colour output, soft proofing and the link between profiles and printing conditions. Fogra describes predictable workflows and objective colour assessment. The International Color Consortium (ICC) promotes open, vendor-neutral colour management. [1-4] Applying these ideas to InkProf, and the potential to reduce test prints, is our own synthesis. It is not a measured saving or an endorsement of InkProf by these organisations.

[1] Adobe - Understanding color management

[2] Adobe - Print with color management in Photoshop

[3] Fogra - Researching Prepress

[4] International Color Consortium - mission

THE WORKFLOW

From colour target to decision

InkProf is a project-based MATLAB app for profiling RGB printers, meaning printers that receive RGB data through their driver. This includes most inkjet photo printers. The app guides the work and preserves the link between inputs and results.

01 Create or open a project

Record the printer, paper, ink, print settings and user in one place. Create or import the RGB target and choose a layout that suits the paper and your instrument.

02 Save TIFF16 and print separately

The app saves 16-bit TIFF files and matching measurement data. Print from your usual application and let the sheets dry. You can close the project and resume later.

03 Measure and review

Measure with a spectrophotometer, such as the i1Pro 2. The app handles the measurement and saves each measurement revision. Review and accept the correct revision before the inputs are locked.

04 Build and check the ICC profile

ArgyllCMS builds the profile. Profiling balances colour accuracy against smooth tonal transitions, so gradients print evenly. Numerical checks examine model fit and colour transformations. A new profile still needs to be assessed with a real print.

05 Verify with a real print

Save the verification target, print it and measure it. The analysis separates the difference from the intended colour from the difference against the profile's prediction. This shows what needs closer investigation.

06 Approve, document or improve

Review the measured result and approve the profile for its stated use, or add colours in another iteration. When improvement levels off, verify the final profile with a fresh control print before delivery. Profiles can also be compared at shared sample points. A delivery without a new control print is clearly labelled numerical- only.

TIFF16 files are saved for separate printing. Measurement takes place in the app. Save the ICC profile and the PDF/HTML certificate wherever you need them; the project keeps the evidence and history.

FEATURES AND DOCUMENTED DELIVERY

Work you can trace

What you need

Licensed MATLAB (base MATLAB is sufficient). Python and ArgyllCMS, both free. A spectrophotometer; the workflow is tested with the i1Pro 2. macOS; the full workflow is not verified on Windows/Linux.

Repeat a result that worked

Each project has its own folder. JSON files record workflow state, settings, iterations and how results relate to each other. The app blocks dependent steps when inputs are missing or have changed. You can open the results log and move the whole project folder between computers.

Spend measurement time where it can help

Add patches based on verification errors, colours in your images or an area you select in the gamut plot. Then print and measure to refine. Paper suggestions adapt to the patch count and your equipment. Compare iterations in a slice at L* = 50 or in 3D.

Make the delivery clear to your client

PDF and HTML reports document the measurements, the current ICC profile and reviewer decisions. ColorChecker verification shows named reference colours beside measured colours with ΔE00, a standard measure of colour difference where lower means closer. The ICC profile, certificate, HTML, evidence and JPG gradient previews are saved in one portable folder. Soft proofing is a model prediction and should not be confused with measured print accuracy.

Standards references supporting assessment

ISO stands for International Organization for Standardization. InkProf documents comparison references, including ISO 12647- 7:2016 via MediaStandard Print 2018, published by bvdm, the German printing and media industry association. Selected limits provide context for assessment. InkProf's own RGB targets do not by themselves constitute a complete standards test or ISO certification.

Open source and responsibility

InkProf is licensed under the GNU GPL version 3 or later. External dependencies have their own licences. The software is provided without warranty; users are responsible for assessing profiles and prints for their intended use. Code, licence and third-party references are on GitHub. See the licence and the certificate's legal appendix for the terms.

MediaStandard Print 2018, Table 30, page 50

InkProf - source and release information

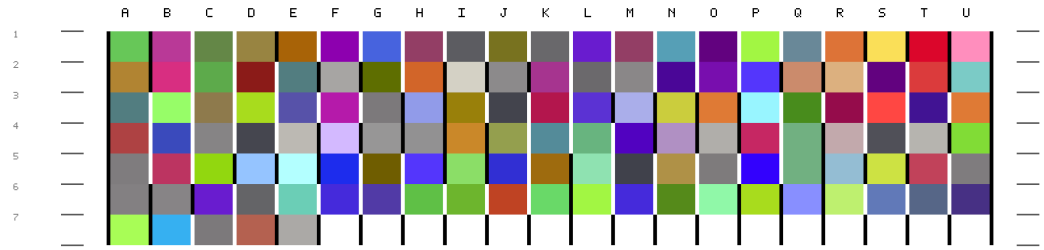
Licensing and third-party references

VISUAL EXAMPLES FROM INKPROF

From colour patches to gamut

A printed colour target provides measurements, and the measurements become a profile. The next page shows how large a colour range that profile describes.

InkProf quality profiling of an RGB inkjet printer



Example target with 131 patches, including repeats, contrast markers and position labels. Do not use this reduced image for measurement.

Measurement and model complement each other

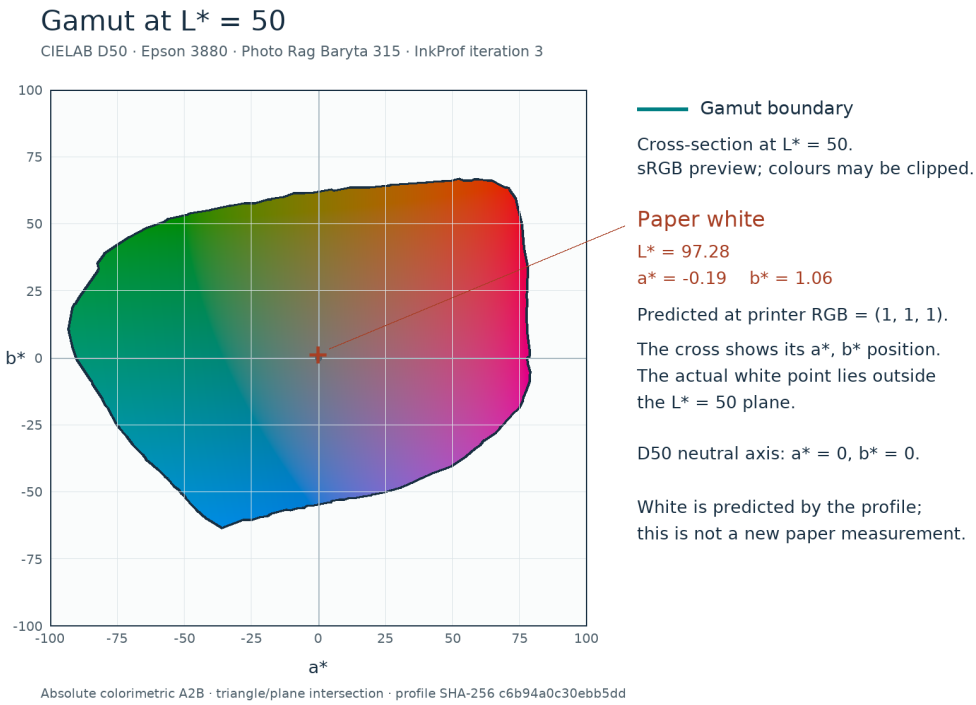
The profile is a model that makes colour conversion possible. Measurement shows what actually reached the paper. Together they support the next decision: use the profile, check a deviating patch or add colours in another iteration.

Existing InkProf example from 2026-09-29. The image is not a new verification result.

ICC GAMUT | CIELAB D50 | L* = 50

The colour gamut at L* = 50

The plot shows the profile colour range at mid-lightness. In each direction the contour indicates how far saturation extends. Colours outside it at this lightness need adjustment during conversion. L* = 50 is midway on the Lab scale from 0 to 100; a* runs from green to red and b* from blue to yellow.



Epson Stylus Pro 3880, Photo Rag Baryta 315, InkProf iteration 3. Absolute colorimetric gamut derived from the ICC profile; colours previewed in sRGB.

Reading the gamut boundary

The outline is where the profile's computed gamut intersects the L* = 50 plane. It shows the colour range at this lightness only, not the entire gamut and not a new measurement.

Paper white is shown separately

The cross marks the profile's predicted paper white, projected onto a*/b*: L* = 97.28, a* = −0.19, b* = 1.06. Its actual lightness therefore lies well above this slice. The value is calculated at printer RGB = (1, 1, 1).

ArgyllCMS A2B, absolute colorimetric, Lab D50. Triangle/plane intersection, no convex hull. Profile SHA-256: c6b94a0c30ebb5dd… Screen colours may be clipped to sRGB.

ArgyllCMS: ICC gamut surface

MEASUREMENT QUALITY AND EARLY CHECKS

Catch errors before they carry forward

InkProf helps you handle measurement errors and check that readings are plausible before they are used to build the profile.

Remeasure a faulty reading

After a reading error, the row can be scanned again. The app tracks confirmed readings and shows which rows remain before the measurement can be saved.

Review plausibility and deviations

Measured values, repeated patches and scan direction help reveal deviations. Results are reviewed before acceptance, so suspicious readings can be investigated or remeasured.

Check the print before a long measurement

New targets have separate red, green and blue control squares. These are measured first and compared with an approved reference for the same printing conditions. Large deviations stop the measurement. They can reveal incorrect colour management, even when an Adobe Photoshop preset was used.

Keep decisions traceable

A completed RGB check with large deviations can be overridden with a written reason, for example when a paper needs investigating. Readings, thresholds and the decision are retained. Older targets without control squares can still be measured.

These checks reduce the risk of carrying faulty prints or readings forward. A passed check must still be followed by profile review and verification.

v1.2.4 | NEW OPTIONS

Understand your print system

Keep using existing instruments

X-Rite has ended updates and support for the i1Pro 2 and earlier i1Pro generations. InkProf communicates with the instrument through ArgyllCMS and is independent of i1Profiler. With compatible versions of MATLAB, Python and ArgyllCMS, everything runs natively on both Apple Silicon and Intel. On Apple Silicon no Rosetta is needed, so the workflow can keep working after Rosetta is removed from macOS. The i1Pro 2 workflow is tested on current macOS; future versions will need verification.

The limits of the print system

Colour accuracy is limited by what the printer, paper, ink and settings can physically reproduce. InkProf helps you assess what is achievable and documents the remaining colour differences. A lack of further improvement does not by itself prove that a physical limit has been reached; the selected profile still needs verification with a measured print.

One project definition

Printer, paper and finish, dye or pigment ink, printer coating and drying time are recorded in one place and included in the measurement certificate.

Correction for optical brighteners

Many papers contain optical brighteners that make them look whiter under UV-rich light. Correction for this can be selected when needed and is documented with the profile and results.

More dark patches for matte paper

An adjustable shadow strategy proposes up to 48 extra dark patches per iteration and denser numerical tables in the shadows. Patch selection follows the current profile's black-to-white range.

Paper to suit your needs

Cut-sheet or roll suggestions adapt to the patch count and the dimensions of your measurement ruler. Suggestions and limits remain editable.

Support throughout the work

The user handbook covers the complete profiling process, from planning and printing to assessment and final verification. Practical guidance helps you decide when to refine, when to stop and how to evaluate the finished result.

Stable feature release

Source release 1.2.1, released 16 October 2026. Saved targets and measurements are preserved. Assess improvements by printing and measuring.

Copyright 2026 Christer Törnkvist. InkProf: GPL-3.0-or-later. External components retain their own licences. Source, notices and validation status are on GitHub.

Open source and downloads

X-Rite: i1Pro 2 EOL

X-Rite: i1Pro 1 support

2026-10-24 | v1.2.4

Identify every printed target

Base target or C2 verification?

Step 3. Measure / select measurement revision reads the base target saved in step 2. A C2 print belongs to C2b. Measure verification print / select saved measurement. If its package is already selected, go directly to that measurement step; do not generate the target again. After all rows are read, choose Save and finish, then C3. Compare model and print and Review feedback.

PROJECT_INFO.txt

The UTF-8 text file retains the exact project name and ID, original project path, workflow step, current cycle/iteration ID, target-origin cycle/iteration ID when found in history, UTC and local copy/creation times, target description when present, source TI2 and its SHA-256, project printing settings and copied filenames with SHA-256. Unknown origin metadata stays unknown.

2026-10-24 | v1.2.4 | b813525

From verification to certificate

C2a prepares the print and C2b measures it

C2a. Prepare verification print (TIFF16) saves or selects the print package. C2b. Measure verification print / select saved measurement reads that sheet. Choose Measure for a new print, not Resume saved session. Save and finish, then run C3. This verifies the existing ICC; it does not build a new profile.

Approval and certificate are separate actions

Selecting a row navigates only. Run Open review and approval, review the results, tick the confirmation and save your assessment. Then select row 14, Save ICC and measurement certificate, and click Save ICC and certificate... Choose Also save copies elsewhere to select the bundle name and location. Save in project only keeps the delivery inside the project. A previous step-19 numerical-only export no longer hides this next step after print approval.

The exact verified ICC is included

The external delivery folder contains <profile>-verified-original.icc, a byte-identical copy of the checked ICC. The normally named ICC has updated name metadata; an optional -v4.4.icc is a converted variant and is not the original verified file. delivery.json records verifiedOriginalICCFile and verifiedOriginalICCSHA256. Keep the whole bundle, including underlag.

1.3.0 | 2026-10-25

Clearer workflow and retained evidence

Review and lock once

Step 4 keeps the measurement preview open until Continue to assessment. B1 validates the selected revision, opens a modal Lock profile input confirmation, then saves a checked snapshot. Activity and elapsed time are shown during validation and saving. Suggested navigation is B1, B2, B3; B2 is shown for recipe review, while Automatic B3 still creates its own recipes.

Correct recorded details

When Project details changes, distinguish Correct recorded details from Changed actual setup. A documentation correction requires a reason and unchanged physical conditions. Original file hashes are checked and a new audit record stores before/after values. Numerical processing changes cannot use this exception. Existing stale steps are not automatically restored.

RGB and other spot measurements

RGB controls, single-patch replacements and paper-white spots use a short start sound for screen-triggered and i1 Pro 2 instrument-button measurements. Calibration does not use the spot start sound. RGB colour rows show page, colour and progress only. The temporary sound-test button has been removed. Page changes are shown inline without a separate confirmation window.

ISSUE-001 closed by user decision 2026-10-25. Included in release 1.3.0.

1.4.0 | 2026-10-25

Measured refinement with preserved originals

Extend latest target

Tick Extend latest target to combine the latest completed gamut print for the exact current ICC with new points. Unprinted proposals are skipped. Previous fitting/development points and supplementary shadow points are retained; exact duplicates and existing training RGB are excluded. Controls are regenerated. Started or paused measurements prevent extending that same target. The earlier TIFF, TI2 and readings remain unchanged.

Set, recalculate, select

Adjust Lab distance (Delta E76), Target per group and Repeat review (Delta E00), then Recalculate. Defaults 1 / 4 / 1 are provisional. Use balancing applies automatic representatives to checked groups; unchecked groups keep all rows. Select all checks/highlights all groups; Clear all restores every row to the proposed input. Neutrals, lightness/RGB extrema and inconsistent repeats stay protected. The target count may therefore be exceeded.

Build a separate candidate

Close only closes the review. Build separate balanced candidate uses the reviewed subset and is disabled when zero points would be omitted. Try different settings and Recalculate rather than forcing removal. Both profiles are compared to ALL original fitting observations, including omitted rows, to available reserved development patches and to matching photographic gradients. These are diagnostics, not final independent print verification. No candidate automatically replaces the active profile.